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***Cimicifuga racemosa*: a systematic review of its clinical efficacy**

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Abstract *Background:* *Cimicifuga racemosa* has long been advocated as an alternative to hormone-replacement therapy. However, recent experimental studies and a clinical trial have raised some questions on its utility. The aim of this systematic review is principally to summarise the trial evidence regarding the efficacy of *C. racemosa* in the treatment of menopausal symptoms. In addition, we also explore the evidence relating to the mechanism of action of this herbal medicine. *Methods:* Searches of seven computerised databases were performed to identify all randomised clinical trials (RCTs) on *C. racemosa* as well as animal and in vitro experiments. No language restrictions were imposed. Data were extracted by both authors according to predefined criteria and are summarised in narrative form.

Results: Four RCTs of *C. racemosa* as a treatment of menopausal symptoms were included. They yielded no compelling evidence for the efficacy of *C. racemosa* on menopausal symptoms. The small beneficial effects observed in some studies could be explained by a central activity, and an oestrogenic activity cannot be completely excluded.

Conclusions: In spite of plausible mechanisms of action of *C. racemosa*, its clinical efficacy for the treatment of menopausal symptoms has not been convincingly demonstrated through rigorous clinical trials. Additional rigorous RCTs and biochemical and chemical investigations are warranted.

Keywords Herbal medicine · *Cimicifuga racemosa* · Black cohosh

Introduction

Menopause is characterised by the cessation of menstruation and by the appearance of many vasomotor, vaginal and psychological symptoms including hot flashes, profuse sweating, flushing of the skin, atrophic vaginitis, irritability and depression. After menopause the risks of coronary heart disease and osteoporosis increase. These effects are linked to hormonal changes, including oestrogen deficiency [1]. Currently, hormone-replacement therapy (HRT) is the most common pharmacological treatment for relief of menopausal symptoms. Unfortunately HRT is associated with endometrial or breast cancer and adverse effects [2, 3, 4, 5]. Therefore, there is a buoyant market for alternative therapies.

The herbal medicine *Cimicifuga racemosa* is often promoted as such an alternative. It is a perennial herb native to North America commonly known as “woman’s herb”. Extracts of the rhizome of *C. racemosa* have long been used for the relief of pain during menses or childbirth and for the treatment of dyspepsia, dysmenorrhea, epilepsy, kidney ailments, malaria, rheumatism and rheumatic pain [6, 7, 8, 9]. Today *C. racemosa* is the principal herbal remedy in for menopausal problems. An ethanolic extract of the rhizome of this plant standardised to contain 1 mg of triterpenes calculated as 27-deoxyactein per 20-mg tablet (trade name Remifemin) is widely marketed for the relief of climacteric (menopausal) disorders including hot flushes and profuse sweating. *C. racemosa* rhizome contains numerous chemical components including triterpene glycosides (actein, 27-deoxyactein, cimicifugoside), phenolic acids (isoferulic acid, fukinolic acid), flavonoids, volatile oils and tannins. The effects of *C. racemosa* are believed to be the result of complex synergistic action of these components. The aim of this systematic review is (i) to

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summarise the evidence for or against the efficacy of *C. racemosa* in providing relief of menopausal symptoms or other conditions and (ii) to summarise our knowledge regarding its possible mechanism of action.

Methods

Computerised literature searches were performed to identify all randomised controlled trials (RCTs) as well as animal and in vitro experiments on *C. racemosa*. Databases included Medline, Embase, Amed, Phytobase, PubMed, CISCOR (Research Council for Complementary Medicine, London) and Cochrane Library (all from their respective inception to July 2001). In addition, several ($n=3$) manufacturers of *C. racemosa* preparations were asked to contribute published or unpublished material, and our own files were hand searched. Bibliographies of the articles thus located were scanned for further relevant publications. No language restrictions were imposed. The search terms used are shown in Table 1.

Clinical trials of *C. racemosa* were included in this review only if performed randomised, blind and controlled. Clinical trials for any indication were included. Animal and in vitro experiments on *C. racemosa* were included only if performed to evaluate its oestrogen-like activity. Studies of this plant in combination products were excluded.

Data were validated and extracted in a standardised, predefined manner (Table 2). The methodological quality of clinical studies was assessed using the scoring system developed by Jadad and colleagues (Table 2) [10].

Results

Several clinical studies examined the effect of *C. racemosa* in menopausal women [11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29]. Only four of these met our inclusion/exclusion criteria [21, 22, 24, 29]. Key data of RCTs are summarised in Table 3. All trials scored at least two of five points on the Jadad score. Three studies reporting positive results in favour of *C. racemosa* scored two, three and two points, respec-

tively [21, 22, 24]. One negative trial scored five points [29]. The trials are described in some detail below. All trials used a commercial product of *C. racemosa*, Remifemin (dosage 40–80 mg).

The first RTC on Remifemin was conducted by Warnecke on 60 women with menopause in 1985 [21]. The women received 40 drops twice daily of *C. racemosa* (80 mg/day), conjugated oestrogens (0.625 mg/day) or diazepam (2 mg/day) for 4 months. The outcome measures were: modified Kupperman menopausal index, self-evaluation depression scale, Hamilton anxiety scale and vaginal epithelium status. *C. racemosa* produced a decrease in climacteric complaints and oestrogen-like stimulation of the vaginal mucosa similar to conjugate oestrogens. Cytological examinations of vaginal epithelium proliferation revealed a similar percentage of eosinophilic epithelial cells with *C. racemosa* and conjugated oestrogens. The women treated with *C. racemosa* extract had a greater percentage of pyknotic nuclei than those on conjugated oestrogens. Greater improvements in neurovegetative symptoms were noted in the experimental group.

A double-blind trial with 80 women aged between 46 years and 58 years compared the effects of a standardised *C. racemosa* extract both with placebo and low-dose oestrogen [22]. The women received two tablets of extract (80 mg), conjugated oestrogens (0.625 mg) or identical placebo twice daily for 3 months. The outcome measures were: Kupperman menopausal index, measuring mainly neurovegetative symptoms, Hamilton anxiety scale for psychological complaints (both recorded every 4 weeks) and proliferation status of vaginal epithelium (recorded at the beginning and the end of the study). At the end of treatment all groups showed improvements compared with baseline. Since the oestrogen dose proved to be too low, only the difference between placebo and *C. racemosa* were evaluated. *C. racemosa* significantly improved all parameters under investiga-

Table 1. Terms used for computer search and number of papers found. *n.r.* not reported, CISCOR Research Council for Complementary Medicine, London

Search terms	Embase Amed	PubMed or Medline	Phytobase	CISCOR	Cochrane Library
<i>Cimicifuga racemosa</i> (botanical name)	70	21	n.r.	8	4
Black cohosh (common English name)	9	18	n.r.	5	1
<i>Actaea racemosa</i> (first botanical name)	4	1	n.r.	2	0
Actein (active ingredient)	3	6	n.r.	1	0
Cimicifugoside (active ingredient)	11	8	n.r.	2	0
Cimicifugic acid (active ingredient)	6	4	n.r.	2	0
Fukinolic acid (active ingredient)	6	4	n.r.	2	0
Remifemin (trade name)	15	6	n.r.	0	0
Cimisan	2	0	n.r.	0	0
Cimicifuga (common Italian name)	120	58	n.r.	21	5
Traubensilberkerze (common German name)	4	0	n.r.	1	0
Wanzenkraut (common German name)	0	0	n.r.	0	0
Black snakeroot (common English name)	0	0	n.r.	0	0
Bugbane (common English name)	0	0	n.r.	0	0
Rattleroot (common English name)	0	0	n.r.	0	0
Rattletop (common English name)	0	0	n.r.	0	0
Rattleweed (common English name)	0	0	n.r.	0	0

Table 2. Jadad scoring system to measure methodological quality of clinical trials

Each 'yes' 1 point; each 'no' 0 points
A. Study described as randomised (includes the use of words such as random, randomly and randomisation)?
B. Study described as double-blind?
C. Description of withdrawals and dropouts?
D. Method of generating the sequence of randomisation described and appropriated (table of random numbers, computer-generated etc.)?
E. Method of double-blinding described and appropriated (identical placebo, active placebo, dummy etc.)?
Deduct 1 point if
F. Method to generate the sequence of randomisation described and inappropriate (patients were allocated alternately, or according to their date of birth, hospital number etc.)
G. Method of double-blinding described and inappropriate (e.g. comparison of tablet versus injection with no double dummy)

tion, compared with placebo. The final score of the menopausal index in women treated with the extract was below 15, which the author considered to be a clinically relevant result.

Lehmann-Willenbrock and Riedel [24] studied the effect of estriol, conjugated oestrogens, oestrogen–gestagen sequential therapy or a *C. racemosa* extract on 60 hysterectomised women who all had at least one intact ovary and complained of climacteric symptoms. The extract (two tablets, 80 mg/day) was administered twice daily for 6 months. The dose of estriol and conjugated oestrogens was 1 mg/day and 1.25 mg/day, respectively, while the dose of oestrogen–gestagen was not specified. The outcome measures were: modified Kupperman index and serum concentrations of follicle-stimulating hormone (FSH) and luteinising hormone (LH). These variables were measured after 4, 8, 12 and 24 weeks. In

Table 3. Clinical studies. *CE* conjugated oestrogens, *EG* oestrogen + gestagen, *FSH* follicle-stimulating hormone, *LH* luteinising hormone

First author, year (reference)	Jadad score	Design	Patients (number)	Dosage of extract	Control treatments (dosage)	Length of treatment (months)	Outcome measures	Main results	Comments
Warnecke, 1985 [21]	2	Randomised, open trial (3 groups)	60 Menopausal women	40 Drops twice daily (80 mg)	CE (0.625 mg/day), diazepam (2 mg/day)	3	Modified Kupperman menopausal index; self-evaluation depression scale; Hamilton anxiety scale; vaginal epithelium status	<i>C. racemosa</i> improved all parameters and increased vaginal epithelium proliferation as well as CE	Open study, no placebo control
Stoll, 1987 [22]	3	Randomised, double-blind trial (3 groups)	80 Women with climacteric complaints	Two tablets twice daily (80 mg)	Placebo, oestrogen (0.625 mg/day)	3	Kupperman menopausal index; Hamilton anxiety scale; vaginal epithelium status	<i>C. racemosa</i> improved all parameters compared with placebo	Lack of oestrogen effects
Lehmann-Willenbrock, 1988 [24]	2	Randomised, open trial (4 groups)	60 Hysterectomised women with at least one intact ovary	Two tablets twice daily (80 mg)	Estriol (1 mg/day), CE (1.25 mg/day), EG	6	Modified Kupperman menopausal index; FSH and LH levels	In all groups Kupperman index decreased. Differences between groups were not significant. Only CE and EG reduced FSH and LH levels	Open study, no placebo group
Jacobson 2001 [29]	5	Randomised, double-blind trial (2 groups)	26 Breast cancer survivors	One tablet twice daily (40 mg)	Placebo	2	Menopausal symptom index; FSH and LH levels	In all groups menopausal symptoms improved. No effect on FSH and LH. No difference between groups	Length of treatment too short

Table 4. Animal and in vitro experiments to investigate the *Cimicifuga racemosa* (CR) mechanism of action. *ER* oestrogen receptor, *PR* progesterone receptor, *LH* luteinising hormone, *FSH* follicle-stimulating hormone, *OPS* ovarian-progesterone secretion, *PYR* pyridinoline, *DPD* deoxypyridinoline, *CP* ceruloplasmin, *PS* prolactin secretion, *n.d.* not described, *n.c.* no comment

Animal experiments			In vitro experiments		
First author, year (reference)	Test(s) used	Main result	First author, year (reference)	Test(s) used	Main result
Gizicky 1944 [30]	Uterine weight, ovary weight	CR increased uterine and ovary weight	Jarry 1985 [33]	ER binding assay	A fraction bound to ER
Foldes 1959 [13]	Uterine weight, induction of oestrus, sedative effect, changes in ovaries and thyroid function	CR increased the uterine weight and induced oestrus	Duker 1991 [25]	ER binding assay	Whole extract did not bind to ER. Some fractions bound to ER
Siess 1960 [31]	Uterine weight, induction of oestrus	No effect was observed	Nesselhut 1993 [42]	Cell proliferation	CR inhibited cell proliferation at concentration > 2.5 µg/ml
Jarry 1985 [32]	FSH levels, LH levels, prolactin levels	A fraction reduced LH levels	Jarry 1996 [43]	OPS	CR inhibited OPS in porcine
Jarry 1985 [33]	LH levels	CHCl ₃ fraction reduced LH levels	Harnischfeger 1996 [44]	ER binding assay, cell proliferation	CR bound to ER, increased the cell proliferation at 1–5 µg/ml and decreased it at > 10 µg/ml
Duker 1991 [25]	LH levels	CR reduced LH levels	Eagon 1996 [45]	ER binding assay	CR did not bind to ER
Einer-Jensen 1996 [34]	Vaginal cornification, uterine weight	CR did not show estrogenic effects	Zava 1998 [46]	ER/PR binding assay, T47D cell proliferation	CR did not bind to ER/PR and did not modify cell proliferation
Eagon, 1997 [35]	Uterine weight, c-myc expression, serum CP levels and hepatic CP mRNA levels	CR root increased uterine weight and CP levels	Lohning 1998 [47]	MCF-7 cell proliferation	CR increased cell proliferation to 0.1–10 µg/ml. 100 µg/ml were ineffective
Eagon 1999 [36]	Uterine weight, LH levels	CR increased uterine weight and decreased LH levels	Lohning 1999 [38]	Basal and TRH stimulated PS	CR reduced both basal and TRH stimulated PS
Jarry 1999 [37]	LH levels, uterine weight and ER α , IgF1, C3, collagen I, osteocalcin expression	CR either reduced LH levels, collagen I and osteocalcin expression or increased ER α expression	Dixon Shanies 1999 [48]	T47D cells proliferation	CR inhibited cell proliferation to 0.1–1 % v/v
Lohning 1999 [38]	Body temperature, ketamine-induced sleeping	CR decreased body temperature and prolonged sleeping time	Eagon 1999 [36]	ER binding assay	CR bound ER
Freudenstein 2000 [39]	Number and size of tumours, FSH, LH, prolactin levels, organ weights	CR did not show oestrogen effects	Jarry 1999 [37]	ER binding assay, luciferase expression in a MCF7- α - and β ER expressing subclone	CR bound to ER and activated the transcription of oestrogen-regulated genes
Nisslein 2000 [40]	PYR and DPD levels, femoral density	CR reduced PYR/DPD levels and bone loss	Freudenstein 1999 [49]	MCF-7 cells proliferation	CR inhibited cells proliferation
Liu 2001 [41]	Days of oestrus, uterine and ovary weight	CR increased the days of oestrus	Liu 2001 [50]	ER α and β binding assay	CR did not show activity
			Liu 2001 [41]	MCF-7 cells proliferation, ER expression	CR increased cells proliferation and ER expression

all groups, a decrease of the modified Kupperman index was observed within 4 weeks after the onset of therapy. There was no significant inter-group difference regarding the Kupperman index at the end of therapy. Only conjugated oestrogens and oestrogen–gestagen sequential therapy significantly modified the serum concentration of FSH and LH.

A recent RCT included 26 female breast cancer survivors [29]. Each woman took one tablet of *C. racemosa* (Remifemin 40 mg) or placebo twice daily with meals for 60 days. At the start and completion of the study, patients completed a detailed menopausal symptom index (heart palpitations, excessive sweating, headaches, poor sleep, depression and irritability or nervousness). The number and intensity of hot flashes were recorded in a diary 3 days before baseline, on days 7–30 and on days 57–60. FSH and LH levels were measured in a subset of patients at both the start and the end of the medication phase. The results showed that both the treatment and placebo groups improved. No statistically significant differences in blood levels of FSH and LH were noted. Except for a reduced number of sweating episodes, no significant beneficial clinical effects of *C. racemosa* compared with placebo were observed.

A total of 14 animal and 15 in vitro studies of *C. racemosa* met our inclusion/exclusion criteria [13, 25, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51]. Key data of all studies are summarised in Table 4. Experimental studies were performed on the capacity of *C. racemosa* to modulate either the plasma hormone levels (LH, FSH), the uterine weight and induction of oestrus in animals or the cell proliferation, the oestrogen-receptor expression and competitive binding assay in vitro. The results fail to yield a uniform picture as the mechanism of action of *C. racemosa*.

Discussion

C. racemosa was approved by the German Commission E for premenstrual discomfort, dysmenorrhea or climacteric (menopausal) neurovegetative ailments. The data summarised above do, however, not conclusively demonstrate its efficacy. Most clinical trials were not well designed. Some were not controlled or randomised and included heterogeneous subject samples [11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 23, 25, 26, 27, 28]. Only four RCTs were located [21, 22, 24, 29]. Three of these trials suggested that an extract of *C. racemosa* had therapeutic benefit on menopausal symptoms improving the Kupperman menopausal index and Hamilton anxiety scale [21, 22, 24]. Trials with active controls suggested that the extract had similar effects to oestrogen replacement therapy [21, 24]. Neither of these studies was designed as an equivalence trial in the strict sense. In the third trial, placebo and oestrogen showed almost the same beneficial effects on the Kupperman index and

Hamilton anxiety scale [22]; and the ineffective oestrogen dosage (0.625 mg/day) used was similar to the dosage (0.625 mg/day) used by Warnecke [21]. It therefore seems possible that the beneficial effect observed by Warnecke [21] and Lehmann-Willenbrock and Riedel [24] was due to a placebo effect.

The recent RCT by Jacobson et al. [29] suggested *C. racemosa* to be ineffective on menopausal symptom index. A large placebo response in menopausal symptoms and no significant difference between the placebo and *C. racemosa* were observed in this RCT. The only significantly reduced number of sweating episodes was, according to the authors, a chance finding. The lack of effect of *C. racemosa* is unlikely to be due to the short duration of therapy (2 months); in a previous RCT, Lehmann-Willenbrock and Riedel [24] reported noticeable benefits of *C. racemosa* within 4 weeks. The lack of rigorous long-term studies is, however, a clear limitation of the evidence available to date.

Early animal studies on an extract of *C. racemosa* had postulated “oestrogen-like” activity as evidenced by an increase in uterine weight and an induction of oestrus [13, 30]. Further studies [25, 32, 33, 37] suggested that *C. racemosa* contained three synergistically acting compounds able to reduce serum LH levels and bind to oestrogen receptors (so as to increase the amount of oestrogen in the blood which decreases the menopause symptoms). Actein and cimicifugoside were believed to be partly responsible for the reduction in LH, while the isoflavone formononetin was thought to bind to oestrogen receptors. However, recent investigations imply that *C. racemosa* contains compounds that act by a mechanism that does not involve oestrogen receptors [39, 46, 50]. The effects seem partly to be due to a central activity mediated by dopaminergic-2 receptors rather than to an oestrogenic activity [38]. Some investigators have found that an extract of *C. racemosa* inhibited the proliferation of human oestrogen receptor-positive breast cancer cells (T47D and MCF-7) [42, 48, 49] and increased oestrogen receptor levels [37, 41]. Since dopaminergic agonists also cause a significant decrease in the proliferation of MCF-7 cells [51] and D1-like dopamine receptors mediate the in vitro transcriptional activation of oestrogen receptors by dopamine [52, 53], the effect of the extract could be due to the presence of dopaminergic substances. This theory is further supported by chemical studies reporting that ethanolic and isopropanolic extracts do not contain the isoflavone formononetin and the flavone kaempferol [54, 55]. These substances are concentrated in the aerial parts of the plant. As some of the early experiments did not use standardised extracts, the detection of formononetin and the effect of *C. racemosa* on oestrogen receptors and LH levels observed in these investigations could be due to contaminants.

In conclusion, the notion that *C. racemosa* has beneficial effects on menopausal symptoms is not supported by the evidence of rigorous RCTs. If an effect on menopausal symptoms exists, it is probably due to an

action discrete from oestrogen receptor activity. Further work is needed to test the efficacy of *C. racemosa* for menopausal symptoms and to assess its long-term effects. In addition, more work is required to identify the pharmacologically active compounds of *C. racemosa* and their actions.

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